3.2 Groups

Definition 3.2.1 (Group)
*A **group** is a **monoid** $(G, \cdot)$ in which every element is invertible.*

*A group G is said to be **abelian** if the binary operation is **commutative**. In this case we typically write the group operation additively*

$$g + h$$

Definition 3.2.2 (Subgroup, Normal Subgroup)
*A **subgroup** $H \leq G$ is a subset with the following properties*

- $e_G \in H$
- $x, y \in H \implies xy \in H$
- $x \in H \implies x^{-1} \in H$

*A subgroup H is said to be **normal** in G if in addition it satisfies*

$$gHg^{-1} \coloneqq \{ghg^{-1} \mid g \in G\} = H$$

for all $g \in G$. NB it is easily verified that in an abelian group every subgroup is normal.

Proposition 3.2.3 (Subgroup is a group)
Let H be a subgroup of $(G, \cdot)$ then $(H, \cdot|_{H \times H})$ is a group.

Example 3.2.4
$\mathbb{Z}$ is an abelian group under addition. The subgroups are of the form $n\mathbb{Z}$.

Definition 3.2.5
*Let $(G, \cdot)$ and $(H, \cdot)$ be groups. A function $\varphi : G \to H$ is a **group homomorphism** if*

- $\varphi(e_G) = e_H$
- $\varphi(x \cdot y) = \varphi(x) \cdot \varphi(y)$

*Define the **image** of φ to be*

$$\mathrm{Im}(\varphi) = \{\varphi(g) \mid g \in G\}$$

*and the **kernel** to be*

$$\ker(\varphi) \coloneqq \{g \in G \mid \varphi(g) = e_H\}$$

It may be verified that $\mathrm{Im}(\varphi)$ is a subgroup of H and $\ker(\varphi)$ is a normal subgroup of G.

Proposition 3.2.6 (Raise to the n-th power)
Let $g \in G$ be a group element. Then there exist a unique group homomorphism

$$g^{(-)} : (\mathbb{Z}, +) \to (G, \cdot)$$

satisfying

$$g^1 = g$$

In other words such that

$$g^0 = e_G$$

$$g^{n+m} = g^n \cdot g^m \quad \forall n, m \in \mathbb{Z}$$

Proposition 3.2.7
Let $g \in G$ be a group element. Then

$$(g^n)^m = g^{nm}$$

for all integers $n, m \in \mathbb{Z}$.

Definition 3.2.8 (Order of an element)
*For $g \in G$ define the **order** of g to be $o(g) \coloneqq \inf\{n \geq 0 \mid g^n = e\}$ where $\inf \varnothing = \infty$.*

*We say g has **finite order** if $o(g) \neq \infty$.*

Definition 3.2.9 (Subgroup generated by an element)
*The **subgroup generated** by an element $g \in G$ is defined to be $\langle g \rangle = \{g^n \mid n \in \mathbb{Z}\}$*

It may be shown that when g has finite order n we have

$$\langle g \rangle = \{e, g, ..., g^{n-1}\}$$

and in particular $\#\langle g \rangle = o(g)$.

Proposition 3.2.10 (Cosets)
Let H be a subgroup of G. The following is an equivalence relation on G

$$g_1 \underset{H}{\sim} g_2 \iff g_1 g_2^{-1} \in H$$

and the equivalence classes are precisely the sets of the form

$$gH = \{gh \mid h \in H\} = [g]_{\underset{H}{\sim}}$$

*for some $g \in G$. Such an equivalence class is called a **right coset** and we denote the set of right cosets by*

$$G/H$$

*Define the **index** of H in G by $[G : H] := \#G/H$. When H is finite each equivalence class has order $\#H$.*

*We say $\{g_i \in G\}_{i \in I}$ is a set of **coset representatives** for H if the corresponding equivalence classes $\{[g_i]\}_{i \in I}$ are pairwise disjoint and cover G.*

Proof. It's trivial to show that $\underset{H}{\sim}$ is an equivalence relation (precisely because H is a subgroup). Therefore by (2.1.6) the equivalence classes form a partition which we denote G/H.

We claim that $[g_1] = g_1 H$. Then $g_2 \in [g_1] \iff g_1 \underset{H}{\sim} g_2 \iff g_2 \underset{H}{\sim} g_1 \iff g_2 g_1^{-1} \in H \iff g_2 \in g_1 H$, which shows that the sets are equal.

The translation map $\psi_g : G \to G$ given by $g' \to gg'$ is bijective (for it has a two-sided inverse equal to $\psi_{g^{-1}}$). So in particular restricts to a bijective map $H \to gH$. This shows that all the cosets have the same order. ∎

Example 3.2.11
$d\mathbb{Z}$ is a subgroup of $\mathbb{Z}$ of index d. A set of coset representatives are $\{0, 1, ..., d-1\}$.

Corollary 3.2.12 (Lagrange's Theorem)
Let $H \leq G$ be a subgroup then

$$\#G = [G : H] \times \#H$$

More generally if $K \leq H$ then

$$[G : K] = [G : H][H : K]$$

Example 3.2.13
$d\mathbb{Z} \subseteq e\mathbb{Z} \iff d \mid e$ and $[e\mathbb{Z} : d\mathbb{Z}] = e/d$.

Proposition 3.2.14
Let $g \in G$ be an element of finite order. Then

$$o(g) \mid \#G$$

Furthermore

$$g^n = e \iff o(g) \mid n$$

Proof. The first statement follows because the order $o(g)$ equals the order of the subgroup $\langle g \rangle$ generated by g.

Let $m = o(g)$ then by the division algorithm $n = qm + r$ for some $r < m$. Then $e = g^n = g^{qm} g^r = (g^m)^q g^r = e^q g^r = g^r$. By minimality we have $r = 0$ and $m \mid n$ as required. ∎

Proposition 3.2.15 (Quotient Group)
Let N be a normal subgroup G. Then the set of cosets

$$G/N$$

forms a group under the binary operation

$$g_1N \cdot g_2N \to (g_1g_2)N$$

with identity eN.

a) There is a canonical surjective group homomorphism

$$\pi : G \to G/N$$

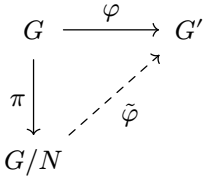
$$g \to gN$$

with kernel N.

b) Let $N \subseteq H$ be a subgroup then define the correseponding subgroup of G/N

$$H/N \coloneqq \pi(H) = \{hN \mid h \in H\}.$$

c) Let $\varphi : G \to G'$ be a homomorphism with $N \subseteq \ker(\varphi)$, then there exists a unique homomorphism $\tilde{\varphi}$ making the diagram commute



such that

a) $\mathrm{Im}(\varphi) = \mathrm{Im}(\tilde{\varphi})$
b) $\ker(\tilde{\varphi}) = \ker(\varphi)/N$

Corollary 3.2.16 (Isomorphism Theorem)
Let $\varphi : G \to H$ be a group homomorphism, then there is a canonical isomorphism

$$G/\ker(\varphi) \xrightarrow{\sim} \mathrm{Im}(\varphi)$$

Proposition 3.2.17 (Correspondence Theorem)
Let $\pi : G \to G'$ be a surjective homomorphism with $\ker(\varphi) = N$ then there is a bijective correspondence of subgroups

$$\begin{aligned} \{H \leq G \mid N \subseteq H\} &\longleftrightarrow \{H' \leq G'\} \\ H &\to \quad \pi(H) \\ \pi^{-1}(H') &\leftarrow \quad H' \end{aligned}$$

which preserves index, that is

$$[G' : H'] = [G : H]$$

Furthermore $\#H' = [H : N]$.

Proposition 3.2.18
Suppose $N \subset G$ and $K \subset N$ are normal subgroups. Then $K \subset G$ is normal and there is a canonical isomorphism

$$(G/N)/(K/N) \xrightarrow{\sim} G/K$$

given by

$$\overline{g+N} \to g + K$$

3.2.1 Cyclic Groups

Definition 3.2.19
*A group G is said to be **cyclic** if there is a surjective group homomorphism*

$$(\mathbb{Z}, +) \longrightarrow (G, \cdot)$$

*equivalently if there is $g \in G$ such that $\langle g \rangle = G$. Such a g is called a **generator** for G.*

Proposition 3.2.20
Consider the additive group $(\mathbb{Z}, +)$. Then

a) Every subgroup is of the form $d\mathbb{Z}$ for $d \geq 0$ and is itself cyclic

b) When $d > 0$, then $Z/d\mathbb{Z}$ has a complete set of coset representatives

$$S := \{0, 1, ..., d-1\}$$

c) In particular $[Z : d\mathbb{Z}] = d$ when $d > 0$

d) $d\mathbb{Z} \subseteq e\mathbb{Z} \iff e \mid d$ and in this case $[e\mathbb{Z} : d\mathbb{Z}] = \frac{d}{e}$

Proof. We prove each in turn

a) By (2.2.6) every subgroup is of the form $d\mathbb{Z}$. Multiplication map $[d] : \mathbb{Z} \to d\mathbb{Z}$ shows it is itself cyclic.
b) By the division algorithm (2.2.5) S is a complete set. Given $i, j \in S$ we note that $|i - j| < d$. And $i \underset{d}{\sim} j \implies d \mid |i - j| \implies |i - j| = 0 \implies i = j$. Therefore the set S consists of distinct coset representatives.
c) This is clear from the previous step
d) The first equivalence is clear. By (3.2.12)

$$[\mathbb{Z} : d\mathbb{Z}] = [\mathbb{Z} : e\mathbb{Z}][e\mathbb{Z} : d\mathbb{Z}]$$

and the result follows.

■

Proposition 3.2.21
Let G be a cyclic group. Then

- *If G is infinite it is isomorphic to $\mathbb{Z}$*
- *If G is finite it is isomorphic to $\mathbb{Z}/n\mathbb{Z}$ for some $n > 0$*

Proof. By the previous Proposition the kernel of the homomorphism $\mathbb{Z} \to G$ is of the form $n\mathbb{Z}$ for $n \geq 0$. By (...) G is isomorphic to $\mathbb{Z}/n\mathbb{Z}$. When $n = 0$ this is canonically isomorphic to $\mathbb{Z}$.

By the previous Proposition $\mathbb{Z}/n\mathbb{Z}$ is finite for $n > 0$ and therefore if G is not finite we must have $n = 0$. Similarly If G is finite then we must have $n > 0$. ■

We analyse the structure of finite cyclic groups in more detail. First recall the definition of **Euler's Totient Function**

Definition 3.2.22 (Euler Totient Function)
Define the function

$$\phi(n) = \#\{0 < d \leq n \mid (d, n) = 1\}$$

Proposition 3.2.23 (Finite Cyclic Groups)
Consider a finite cyclic group G of order n. Then

a) The order of g^r is $\frac{n}{n,r}$ where $0 < r \leq n$.
b) There are $\phi(n)$ generators
c) For every $d \mid n$ there is a unique subgroup of order $\frac{n}{d}$ given by $\langle g^d \rangle$, which is cyclic.
d) For $d \mid n$ there are precisely $\phi(d)$ elements of order d
e) There are precisely d elements of order dividing d

Proof. We prove each in turn

a) $(g^r)^s = e_G \iff g^{rs} = e_G \overset{(3.2.14)}{\iff} n \mid rs \overset{(2.2.13)}{\iff} \frac{n}{n,r} \mid s$. Therefore g^r has order $\frac{n}{n,r}$ as required.
b) Note h is a generator iff $o(h) = n$. So g^r is a generator iff $(n, r) = 1$ by the previous step. As $G = \{g, g^2, ..., g^n\}$ the result follows by definition of the totient function.
c) Recall there is a canonical surjective morphism $\pi : \mathbb{Z} \to G$ with kernel $n\mathbb{Z}$ and $\pi(1) = g$. By (3.2.17) the subgroups H of G correspond bijectively to subgroups H' of $\mathbb{Z}$ containing $n\mathbb{Z}$, preserving the index. By (3.2.20) these are of the form $H' = d\mathbb{Z}$ for $d \mid n$, which correspond under π to subgroups $H = \langle g^d \rangle$. Further $[G : \langle g^d \rangle] = [\mathbb{Z} : d\mathbb{Z}] = d$ whence $\#\langle g^d \rangle = \frac{n}{d}$. By definition $\langle g^d \rangle$ is cyclic.
d) Let $G[d]$ be the unique (cyclic) subgroup of order d. If h has order d then $\langle h \rangle$ has order d, and therefore by uniqueness is equal to $G[d]$. In particular $h \in G[d]$. Therefore by the previous part there are $\phi(d)$ elements of order d

e) Suppose h has order $e \mid d$. Both G and $G[d]$ contain a unique subgroup of order e and therefore by uniqueness this is simply $G[e] \subseteq G[d]$. Similarly by uniqueness $G[e] = \langle h \rangle$. Therefore $h \in G[d]$. Conversely suppose $h \in G[d]$ then $o(g) \mid d$ by (3.2.14). Therefore $G[d]$ consists of all the elements of order dividing d.

■

Corollary 3.2.24
Let n be a positive integer then

$$n = \sum_{d|n} \phi(d)$$

Proof. Consider a cyclic group G of order n. Every element has order dividing n so the result follows from the previous Proposition by partitioning the group G into subsets consisting of elements of equal order. ■

For an abelian group G define the following subgroup

$$G[d] := \{g \in G \mid g^d = e\}.$$

We have shown for a cyclic group that $\#G[d] = d$ whenever $d \mid n$ and it is empty otherwise. We claim that this can be used to characterize cyclic groups. NB the following is adapted from this stackexchange answer

Proposition 3.2.25 (Characterization of cyclic group)
Let G be a finite abelian group such that $\#G[d] \leq d$ for all $d \mid n$. Then G is cyclic.

Proof. Let $n = \#G$ and G_d be the subset of elements of order exactly d. Then we wish to show that G_n is non-empty as any element of this set will be a generator. We actually show that $\#G_d = \phi(d) > 0$ whenever $d \mid n$.

Note that $G_d \subseteq G[d]$. If it's non-empty then for any $y \in G_d$, we have $\langle y \rangle$ is a subgroup of $G[d]$ of order d. As $\#G[d] \leq d$ we have $G[d] = \langle y \rangle$ is cyclic of order d. In other words G_d is equal to the set of generators for $G[d]$. By the previous Proposition $G[d]$ has $\phi(d)$ generators. We conclude that for all $d \mid n$ we have G_d is either empty or of order $\phi(d)$.

Therefore

$$n = \sum_{d|n} \#G_d \leq \sum_{d|n} \phi(d) = n$$

Therefore we must have equality everywhere and $\#G_d = \phi(d)$ as required. ■

Example 3.2.26
Let $G = \mathbb{Z}/p\mathbb{Z} \times \mathbb{Z}/p\mathbb{Z}$ of order p^2. Then $G[p] = G$ so $\#G[p] = p^2 > p$.

3.2.2 Group Actions

Definition 3.2.27 (Group Action)
Let G be a group and S a set. A group action of G on S is a map

$$G \times S \longrightarrow S$$

$$(g, s) \longrightarrow g \cdot s$$

such that

- $es = s$
- $g(hs) = (gh)s$

Definition 3.2.28 (Faithful group action)
A group action G on S is faithful if

$$gs = s \quad \forall s \in S \Longrightarrow g = e$$

Definition 3.2.29 (Free group action)
A group action G on S is free if

$$g \neq e \Longrightarrow gs \neq s \quad \forall s$$

Definition 3.2.30 (Orbit/Stabilizer)
Let G be a group with an action on S and $s \in S$. Define the stabilizer subgroup

$$G_s := \{g \in G \mid gs = s\}$$

and the orbit

$$Gs := \{gs \mid g \in G\}$$

Proposition 3.2.31 (S is disjoint union of orbits)
Let G be a group with an action on S. Then the following is an equivalence relation

$$s \sim t \iff gs = t \text{ some } g \in G$$

and the equivalence classes are precisely the orbits of elements of S under G. Further S is the disjoint union of orbits.

Remark 3.2.32
An action is free if and only if $G_s = \{e\}$ for all $s \in S$.

Proposition 3.2.33 (Orbit-Stabilizer Theorem)
Let G be a group with an action on S. Given an element $s \in S$ there is a natural bijection

$$G/G_s \longrightarrow Gs$$

between the cosets of G_s and the orbit Gs. In particular when G is finite

$$\#G = \#(Gs) \times \#G_s$$

and when the action is free

$$\#G = \#(Gs)$$

3.2.3 Symmetric Group

Definition 3.2.34 (Symmetric Group)
Let S_n denote the set of permutations (bijections) of $\{1, ..., n\}$.

*Permutations $\sigma, \tau \in S_n$ are called **disjoint** if the supports are disjoint. Note disjoint permutations commute.*

Definition 3.2.35 (Cycle)
Let $i_1, ..., i_r \in J_n$ be an ordered r-tuple, the permutation which maps

$$i_k \to \begin{cases} i_{k+1} & k < r \\ i_1 & k = r \end{cases}$$

*is denoted by $(i_1\, i_2 \ldots i_r)$ and called a **cycle**.*

*A cycle with two elements $(i\, j)$ is called a **transposition**. Finally an **adjacent transposition** is one of the form $(i\, i+1)$.*

Proposition 3.2.36
Let $\sigma \in S_n$. Then σ may be represented by

a) a product of disjoint cycles, which is unique up to permutation of cycles.
b) a product of transpositions, the number of which is unique modulo 2
c) There is a well-defined group homomorphism

$$\varepsilon : S_n \to \{-1, 1\}$$

such that

$$\varepsilon((i\, j)) = -1$$

and $\varepsilon(1) = 1$.
d) A cycle σ of length r satisfies $\varepsilon(\sigma) = (-1)^{r+1}$

3.2.4 Shuffle Permutations

Definition 3.2.37
*Let $p + q = n$ be positive integers. We define the subgroup of **shuffle permutations***

$$\mathrm{Sh}(p, q) := \{\sigma \in S_n \mid \sigma(1) < ... < \sigma(p) \text{ and } \sigma(p+1) < ... < \sigma(p+q)\}$$

Similarly if $p+q+r=n$ are positive integers define the subgroup

$$\mathrm{Sh}(p,q,r) \coloneqq \{\sigma \in S_n \mid \sigma(1) < ... < \sigma(p) \text{ and } \sigma(p+1) < ... < \sigma(p+q) \text{ and } \sigma(p+q+1) < ... < \sigma(p+q+r)\}$$

Let $u : [1,p] \to [1,n]$ and $v : [1,q] \to [1,n]$ be injective maps such that $u([1,p]) \cap v([1,q]) = \varnothing$. Define $u \star v \in S_n$ by

$$(u \star v)(i) \coloneqq \begin{cases} u(i) & 1 \leq i \leq p \\ v(i-p) & p < i \leq p+q \end{cases}$$

If u, v are order preserving then $u \star v \in \mathrm{Sh}(p,q)$. Every shuffle permutation is of this form.

Lemma 3.2.38
Let $X, Y \subset \mathbb{N}$ be a finite sets of order n. There exists a unique order isomorphism

$$u : X \to Y$$

Proposition 3.2.39
Let $p+q=n$ be positive integers. There is a bijection

$$\mathrm{Sh}(p,q) \to \{u : [1,p] \to [1,n] \text{ order preserinvg}\}$$

$$\sigma \to \sigma|_{[1,p]}$$

Proof. We provide an explicit inverse. For given u the set $Y \coloneqq [1,n] \setminus u([1,p])$. As u is order preserving it is injective, whence $\#Y = n - \#u([1,p]) = n-p = q$. Therefore there is an order isomorphism $v : [1,q] \to Y$ and we may define the shuffle permutation $\sigma \coloneqq u \star v$. ∎

Proposition 3.2.40
Let $p+q=n$ be positive integers. There is an injective group homomorphism

$$\begin{aligned} S_p \times S_q \to & \qquad S_n \\ (\sigma_1, \sigma_2) \to & \ \sigma_1 \star (\sigma_2 + p) \end{aligned}$$

The image is the set $\sigma \in S_n$ for which $\sigma([1,p]) = [1,p]$ (equivalently $\sigma([p+1,q]) = \sigma([p+1,q])$.

Proposition 3.2.41
Let $p+q=n$ be positive integers. There is a bijection

$$\mathrm{Sh}(p,q) \to S_n / \left(S_p \times S_q\right)$$

Furthermore $\#\,\mathrm{Sh}(p,q) = \frac{(p+q)!}{p!q!}$. By a similar argument $\#\,\mathrm{Sh}(p,q,r) = \frac{(p+q+r)!}{p!q!r!}$.

Proof. Let $\sigma \in S_n$. Then there exists an order isomorphism $u : [1,p] \to \sigma([1,p])$ and $v : [1,q] \to \sigma([p+1,p+q])$. Define $\sigma_1(i) \coloneqq u^{-1}(\sigma(i))$ and $\sigma_2(j) \coloneqq v^{-1}(\sigma(j+p))$. Then we may verify that $\sigma_1 \in S_p$, $\sigma_2 \in S_q$ and

$$(u \star v) \circ (\sigma_1 \star (\sigma_2 + p)) = \sigma$$

so the map is surjective.

Suppose $\sigma, \sigma' \in \mathrm{Sh}(p,q)$ are such that

$$\sigma = \sigma' \circ (\sigma_1 \star (\sigma_2 + p))$$

for $\sigma_1 \in S_p$ and $\sigma_2 \in S_q$. If $\sigma_1 \neq e$ then there exists $1 \leq i < j \leq p$ such that $\sigma_1(j) < \sigma_1(i)$. Then by monotonicity of σ' we have $\sigma(j) = \sigma'(\sigma_1(j)) < \sigma'(\sigma_1(i)) = \sigma(i)$, which contradicts monotonicity of σ. So we conclude $\sigma_1 = e$ and similarly $\sigma_2 = e$. Therefore $\sigma = \sigma'$, and the map is injective. By the Orbit-Stabilizer theorem we have

$$\#\,\mathrm{Sh}(p,q) = \#S_n / \#\left(S_p \times S_q\right)$$

from which the result follows. ∎

Definition 3.2.42 (Circulant Permutation)
Let n be an integer, and write $\rho_n \in S_n$ for the permutation

$$\{1, ..., n\} \to \{2, 3, ..., n, n+1\}$$

Note that $\varepsilon(\rho_n) = (-1)^{n+1}$.

Proposition 3.2.43 (Symmetry of Shuffling)
Let $p+q=n$ be positive integers. Then there is a bijection

$$\begin{aligned} \mathrm{Sh}(p,q) &\to \mathrm{Sh}(q,p) \\ \sigma &\to \sigma \circ \rho^p_{p+q} \end{aligned}$$

Note that $\varepsilon\left(\rho^p_{p+q}\right) = (-1)^{pq}$ *by (3.2.42).*

Proposition 3.2.44 (Associativity of Shuffling)
The following map is a bijection

$$\begin{aligned} \pi : \mathrm{Sh}(p+q,r) \times \mathrm{Sh}(p,q) &\to \mathrm{Sh}(p,q,r) \\ (\sigma_1,\sigma_2) &\to \sigma_1 \circ (\sigma_2 \star (1_r + p + q)) \end{aligned}$$

Similarly so is

$$\begin{aligned} \pi' : \mathrm{Sh}(p,q+r) \times \mathrm{Sh}(q,r) &\to \mathrm{Sh}(p,q,r) \\ (\sigma_1,\sigma_2) &\to \sigma_1 \circ \left(1_p \star (\sigma_2 + p)\right) \end{aligned}$$

Proof. It is straightforward to verify that $\pi(\sigma_1,\sigma_2)$ is injective and satisfies the ordering properties. Therefore π is well-defined and we claim it is surjective.

Given $\sigma \in \mathrm{Sh}(p,q,r)$ let $u : [1,p+q] \to \sigma([1,p+q])$, $v : [1,r] \to \sigma([p+q+1,p+q+r])$ be the unique order isomorphisms. Define $\sigma_1 = u \star v$ and $\sigma_2 := u^{-1} \circ \sigma|_{[1,p+q]}$. We may verify that $\sigma_1 \in \mathrm{Sh}(p+q,r)$ and $\sigma_2 \in \mathrm{Sh}(p,q)$, and furthermore that $\pi(\sigma_1,\sigma_2) = \sigma$. By counting π is injective.

For the second case let $u : [1,p] \to \sigma([1,p])$, $v : [1,q+r] \to \sigma([p+1,p+q+r])$ be the unique order isomorphisms. Define $\sigma_1 = u \star v$ and $\sigma_2(j) := v^{-1}(\sigma(j+p))$. The verification follows as before. ∎

3.2.5 Totally Ordered Abelian Group

Definition 3.2.45 (Ordered Abelian Group)
An abelian group $(G,+)$ *together with a total order* $\leq$ *is an* ***ordered abelian group*** *if it satisfies*

$$x \leq y \implies x + z \leq y + z \quad \forall x,y,z \in G$$

Define $G^+ := \{g \in G \mid (0 \leq g) \wedge (g \neq 0)\}$ *and* $G^- := \{g \in G \mid (g \leq 0) \wedge (g \neq 0)\}$.